

Consumer Retail Analysis

Project Report & Dashboard Insights

This project analyzes 3,900 customer shopping transactions to understand purchasing patterns, customer behavior, product performance, subscription adoption, and service preferences.

Project Snapshot

Dataset	Customers	Avg. Purchase	Avg. Rating
3,900 transactions	3.9K	\$59.76	3.75 / 5

1. Project Objective

The objective is to use consumer shopping data to identify trends and customer patterns that can support better marketing, product, subscription, and service decisions.

2. Data Preparation

The source project describes a Python-based preparation process using pandas. It includes dataset inspection, missing-value handling for Review Rating, column standardization, feature engineering, consistency checks, and loading the cleaned data into PostgreSQL for SQL analysis.

3. Key SQL Findings

Area	Finding
Revenue by Gender	Male customers: \$157,890; Female customers: \$75,191.
Discount Users	839 customers used discounts while spending above the average purchase amount.
Top Rated Products	Gloves 3.86, Sandals 3.84, Boots 3.82, Hat 3.80, Skirt 3.78.
Shipping Spend	Express average purchase: \$60.48; Standard: \$58.46.
Subscriptions	1,053 subscribers and 2,847 non-subscribers.
Customer Segments	3,116 Loyal, 701 Returning, 83 New.

4. Power BI Dashboard

The dashboard below is the actual dashboard supplied for this project. It is included directly in the report so the written analysis can be compared with the visual output.

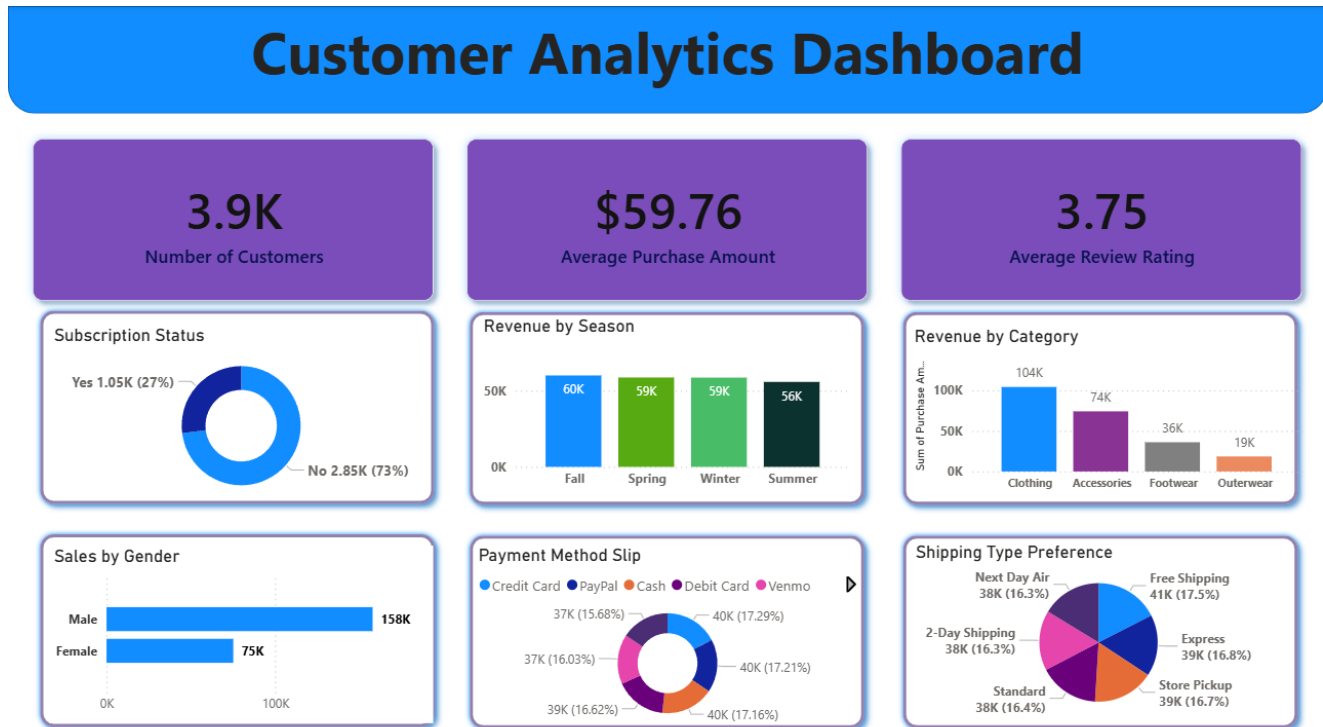


Figure 1. Customer Analytics Dashboard — supplied project dashboard.

Dashboard Reading

Visual	What the dashboard shows
KPI Cards	3.9K customers, \$59.76 average purchase amount, and 3.75 average review rating.
Subscription Status	1.05K / 27% Yes and 2.85K / 73% No.
Revenue by Season	Fall 60K, Spring 59K, Winter 59K, Summer 56K.
Revenue by Category	Clothing 104K, Accessories 74K, Footwear 36K, Outerwear 19K.
Sales by Gender	Male 158K and Female 75K.
Payment Method	Credit Card, PayPal, Cash, Debit Card and Venmo are shown with closely distributed values.
Shipping Preference	Free Shipping 41K, Express 39K, Store Pickup 39K, Standard 38K, 2-Day Shipping 38K, Next Day Air 38K.

5. Business Insights & Recommendations

The recommendations below are based on the SQL findings and the supplied dashboard.

1. Improve Subscription Conversion

The dashboard shows 73% of customers as non-subscribers. Repeat buyers also represent a large conversion opportunity. Targeted subscription benefits can be tested for customers with established purchase activity.

2. Focus on High-Revenue Categories

Clothing is the leading category in the dashboard at approximately 104K, followed by Accessories at 74K. Inventory and promotional planning can prioritize these stronger categories.

3. Use Seasonal Demand

Fall is the strongest season shown at about 60K. Marketing campaigns and inventory preparation can be planned ahead of the stronger seasonal period.

4. Use Customer Segmentation

The SQL analysis identifies 3,116 Loyal customers, 701 Returning customers, and 83 New customers. Different engagement strategies can be designed for each segment.

5. Review Discount Dependency

Hat, Sneakers, Coat, Sweater, and Pants have the highest discount rates in the SQL analysis. A targeted rather than blanket discount strategy can help protect margins.

6. Promote Highly Rated Products

Gloves, Sandals, Boots, Hat, and Skirt have the highest average ratings in the SQL findings. These products can be considered for stronger merchandising and promotional placement.

7. Use Shipping Preferences

The dashboard shows relatively balanced shipping preferences. Shipping offers can therefore be tested as a customer-experience and conversion lever.

6. Tools Used

Python • Pandas • PostgreSQL / SQL • Power BI • GitHub

7. Conclusion

The project combines Python data preparation, SQL-based analysis, and Power BI visualization to provide a practical view of customer shopping behavior. The dashboard gives stakeholders a concise way to monitor customer volume, spending, ratings, subscription adoption, seasonal performance, category revenue, gender-based sales, payment methods, and shipping preferences.